

$N \times$ DiT block ($L=12$)

per-object token

state $x_\tau = (u, v, \cos \theta, \sin \theta)$

lean cond: log-size, ref-pose, motif-head

learned category embedding

Linear $\rightarrow h$

flat tokens (no motif hierarchy)

adaLN(c) $\cdot$ scale/shift

Multi-Head Self-Attention

+ geo-bias + RelHead bias s_{ij}

per-object wall feature

nearest-wall dist $\cdot$ normal misalign

adaLN(c)

MLP (GELU)

residual $\cdot$ repeated $\times L$

τ time-embed

+ $g(P_t)$, room, z_{style}

RelHead

s_{ij} bias (pos+orient)

out adaLN + heads

pose $\cdot$ wall $\cdot$ ortho $\cdot$ keep

Euler ODE 50 steps

$x \leftarrow x + d\tau (v_\theta - \text{ramp} \cdot \nabla \Phi)$

from prior x_0

flow in world frame

iterate $\tau: 0 \rightarrow 1$

$c = \text{global}(\tau, g(P_t), \text{room}, z_{\text{style}})$